Ps 07

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1 Problem 1

1.1 Task

Let $f : (-\frac{1}{2}, \infty) \times (-\frac{1}{2}, \infty) \to \mathbb{R}$ be defined by

$$f(x, y) = \log(1 + x + y)$$

(a) Determine the Taylor polynomial of order 3 of f at $(0, 0)$. (b) Use your result to approximate $f(0.1, 0.1)$. (c) Plot heatmaps of f, of the Taylor polynomial of order 3, and of the difference of the two in $[0, 1]^2$.

1.2 Solution

1.2.1 (a)

$$f(x) = \sum_{|\alpha| \leq K} \frac{\partial^\alpha f(x_0)}{\alpha!}(x - x_0)^\alpha + o(\|x - x_0\|^K)$$

Since $f(x, y) = \log(1 + x + y)$ depends on (x, y) only through $s = x + y$, all partials of equal total order coincide:

$$\begin{aligned}
\partial_1 f = \partial_2 f &= \frac{1}{1 + x + y} \\
\partial_1^2 f = \partial_1 \partial_2 f = \partial_2^2 f &= -\frac{1}{(1 + x + y)^2} \\
\partial_1^3 f = \partial_1^2 \partial_2 f = \partial_1 \partial_2^2 f = \partial_2^3 f &= \frac{2}{(1 + x + y)^3}
\end{aligned}$$

Evaluated at $(0, 0)$:

$$\begin{aligned}
f(0, 0) &= 0 \\
\partial_1 f(0, 0) &= 1 \\
\partial_1^2 f(0, 0) &= -1 \\
\partial_1^3 f(0, 0) &= 2
\end{aligned}$$

Each term of P_3 has the form

$$\text{term}_\alpha = \frac{\partial^\alpha f(0,0)}{\alpha!} x^{\alpha_1} y^{\alpha_2}.$$

Order 0: $\alpha = (0,0)$, term $= 0$.

Order 1:

- $\alpha = (1,0)$: $\dfrac{1}{1!\,0!}\,x = x$
- $\alpha = (0,1)$: $\dfrac{1}{0!\,1!}\,y = y$

Sum: $x + y$.

Order 2:

- $\alpha = (2,0)$: $\dfrac{-1}{2!\,0!}\,x^2 = -\dfrac{1}{2}x^2$
- $\alpha = (1,1)$: $\dfrac{-1}{1!\,1!}\,xy = -xy$
- $\alpha = (0,2)$: $\dfrac{-1}{0!\,2!}\,y^2 = -\dfrac{1}{2}y^2$

Sum: $-\dfrac{1}{2}x^2 - xy - \dfrac{1}{2}y^2 = -\dfrac{1}{2}(x+y)^2$.

Order 3:

- $\alpha = (3,0)$: $\dfrac{2}{3!\,0!}\,x^3 = \dfrac{1}{3}x^3$
- $\alpha = (2,1)$: $\dfrac{2}{2!\,1!}\,x^2y = x^2y$
- $\alpha = (1,2)$: $\dfrac{2}{1!\,2!}\,xy^2 = xy^2$
- $\alpha = (0,3)$: $\dfrac{2}{0!\,3!}\,y^3 = \dfrac{1}{3}y^3$

Sum: $\dfrac{1}{3}x^3 + x^2y + xy^2 + \dfrac{1}{3}y^3 = \dfrac{1}{3}(x+y)^3$.

Hence,

$$P_3(x,y) = (x+y) - \frac{1}{2}(x+y)^2 + \frac{1}{3}(x+y)^3.$$

1.2.2 (b)

$$
\begin{aligned}
P_3(0.1, 0.1) &= (0.1 + 0.1) - \frac{1}{2}(0.1 + 0.1)^2 + \frac{1}{3}(0.1 + 0.1)^3 \\
&= (0.2) - \frac{1}{2}(0.2)^2 + \frac{1}{3}(0.2)^3 \\
&= 0.182\dot{6} \\
\log(1.2) &\approx 0.18232
\end{aligned}
$$

1.2.3 (c)

```
using Plots

f(x, y) = log(1 + x + y)
P3(x, y) = (x + y) - 1/2 * (x + y)^2 + 1/3 * (x + y)^3
err(x, y) = f(x, y) - P3(x, y)

xs = 0:0.01:1
ys = 0:0.01:1

cl = (0, P3(1, 1))

p1 = heatmap(xs, ys, f,
             title="f",
             clim=cl,
             aspect_ratio=1,
             c=:viridis)

p2 = heatmap(xs, ys, P3,
             title="P3",
             clim=cl,
             aspect_ratio=1,
             c=:viridis)

p3 = heatmap(xs, ys, err,
             title="f - P3",
             aspect_ratio=1,
             c=cgrad(:reds, rev=true))

p4 = surface(xs, ys, f,
             title="f vs P3",
             c=:viridis,
             alpha=0.7)
```

```
surface!(p4, xs, ys, P3,
         c=:viridis,
         alpha=0.7)

plot(p1, p2, p3, p4, layout=(2,2), size=(1200, 1200))

mkpath("./assets")
savefig("./assets/p_01_c.png")
```

2 Problem 2

2.1 Task

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = x^4 + y^4 - 2x^2 - 2y^2$$

(a) Compute the gradient $\nabla f(x, y)$. (b) Determine all critical points of f and classify them as local minima, local maxima, or saddle points via the Hessian matrix. (c) Implement gradient descent for f in Julia. Experiment with several starting values and describe which critical points the algorithm converges to.

2.2 Solution

2.2.1 (a)

$$\nabla f(x, y) = \begin{pmatrix} 4x^3 - 4x \\ 4y^3 - 4y \end{pmatrix}$$

2.2.2 (b)

$$\begin{aligned} \nabla f(x, y) &= \begin{pmatrix} 4x(x^2 - 1) \\ 4y(y^2 - 1) \end{pmatrix} = \begin{pmatrix} 4x(x + 1)(x - 1) \\ 4y(y + 1)(y - 1) \end{pmatrix} \\ \nabla f(x, y) = 0 &\iff x \in \{-1, 0, 1\} \text{ and } y \in \{-1, 0, 1\}. \end{aligned}$$

Hence, 9 critical points:

$$P = \{-1, 0, 1\} \times \{-1, 0, 1\} = \left\{ \begin{array}{ccc} (-1, -1), & (0, -1), & (1, -1) \\ (-1, 0), & (0, 0), & (1, 0) \\ (-1, 1), & (0, 1), & (1, 1) \end{array} \right\}.$$

$$\partial_1 f = 4x^3 - 4x \qquad \partial_2 f = 4y^3 - 4y$$
$$\partial_{11} f = 12x^2 - 4 \qquad \partial_{22} f = 12y^2 - 4$$
$$\partial_{21} f = 0 \qquad \partial_{12} f = 0$$

$$H_f(x, y) = \begin{pmatrix} 12x^2 - 4 & 0 \\ 0 & 12y^2 - 4 \end{pmatrix}$$

Hence,

$$H_f(\pm 1, \pm 1) = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix} \quad \text{(local min)}$$
$$H_f(0, 0) = \begin{pmatrix} -4 & 0 \\ 0 & -4 \end{pmatrix} \quad \text{(local max)}$$
$$H_f(\pm 1, 0) = \begin{pmatrix} 8 & 0 \\ 0 & -4 \end{pmatrix} \quad \text{(saddle)}$$
$$H_f(0, \pm 1) = \begin{pmatrix} -4 & 0 \\ 0 & 8 \end{pmatrix} \quad \text{(saddle)}$$

2.2.3 (c)

$f(x, y) = g(x) + g(y)$ with $g(t) = t^4 - 2t^2$, so the gradient separates and each axis evolves independently. Generic starts give $x_k \to \text{sign}(x_0)$, $y_k \to \text{sign}(y_0)$, hence GD converges to a corner $(\pm 1, \pm 1)$. Saddles and the origin require $x_0 = 0$ or $y_0 = 0$ — measure zero.

```
using Plots

f(x, y) = x^4 + y^4 - 2x^2 - 2y^2
f(x, y) = [4x^3 - 4x, 4y^3 - 4y]

function gd(x0; =0.01, N=1000)
    path = zeros(2, N);
    path[:, 1] = x0
    for k in 1:N-1
        path[:, k+1] = path[:, k] .-  .* f(path[1, k], path[2, k])
    end
    return path
end

starts = [[ 1.5,  1.7],
```

```
          [-1.5,  1.7],
          [-1.5, -1.7],
          [ 1.5, -1.7],
          [ 1.2,  0.05],
          [-0.05, -1.2],
          [ 1.5,  0.0]]

xs = -2:0.05:2
ys = xs

contour(xs, ys, f, levels=50, c=:viridis, aspect_ratio=1, colorbar=false, size=(800, 800))

for s in starts
    p = gd(s)
    plot!(p[1,:], p[2,:], lw=2, label=false)
    scatter!([p[1,1]],   [p[2,1]],   c=:black, msw=0, label=false)
    scatter!([p[1,end]], [p[2,end]], c=:black, msw=0, label=false)
end

mkpath("./assets")
savefig("./assets/p_02_c.png")
```

```
/Users/ziro2173/personal/mfds/01_bachelor/02_semester/01_analysis/02_exercises/ps_07/assets/p_02
```

3 Problem 3

3.1 Task

Let V and W be normed spaces and let $f_1, \ldots, f_n : V \to \mathbb{R}$ be convex. Prove that the following functions are also convex.

(a) $\sum_{i=1}^{n} \alpha_i f_i$ for $\alpha_1, \ldots, \alpha_n \geq 0$. (b) $g : V \to \mathbb{R}$ given by $g(x) = \max_{i=1,\ldots,n} f_i(x)$. (c) If $A : W \to V$ is linear and $b \in V$, then $g : W \to \mathbb{R}$ given by

$$g(x) = f_1(Ax - b)$$

is also convex.

3.2 Solution

Definition 4.3.19 (convex function). A function $f : \mathbb{R} \to \mathbb{R}$ is called a convex if for all $x, y \in \mathbb{R}$ and all $t \in [0, 1]$,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

3.2.1 (a)

Let $h = \sum_{i=1}^{n} \alpha_i f_i$ for $\alpha_1, \ldots, \alpha_n \geq 0$.

By convexity of each f_i:

$$f_i(tx + (1-t)y) \leq tf_i(x) + (1-t)f_i(y)$$

Since $\alpha_i \geq 0$, multiplying preserves the inequality:

$$\begin{aligned}
\sum_{i=1}^{n} \alpha_i f_i(tx + (1-t)y) &\leq \sum_{i=1}^{n} \alpha_i t f_i(x) + \sum_{i=1}^{n} \alpha_i (1-t) f_i(y) \\
\sum_{i=1}^{n} \alpha_i f_i(tx + (1-t)y) &\leq t \sum_{i=1}^{n} \alpha_i f_i(x) + (1-t) \sum_{i=1}^{n} \alpha_i f_i(y) \\
h(tx + (1-t)y) &\leq th(x) + (1-t)h(y)
\end{aligned}$$

3.2.2 (b)

Let $g(x) = \max_{i=1}^{n} f_i(x)$.

By definition of g, $\forall i$:

$$f_i(x) \leq g(x), \qquad f_i(y) \leq g(y)$$

Since $t, (1-t) \geq 0$, inequality preserves:

$$tf_i(x) + (1-t)f_i(y) \leq tg(x) + (1-t)g(y)$$

Chaining with the convexity of f_i:

$$\begin{aligned}
f_i(tx + (1-t)y) &\leq tf_i(x) + (1-t)f_i(y) \\
&\leq tg(x) + (1-t)g(y)
\end{aligned}$$

This holds $\forall i \in \{1, \ldots, n\}$, and the right-hand side is independent of i. Taking the max over i on the left:

$$g(tx + (1-t)y) = \max_{i=1,\dots,n} f_i(tx + (1-t)y) \le tg(x) + (1-t)g(y)$$

3.2.3 (c)

Let $g : W \to \mathbb{R}$, $g(x) = f_1(Ax - b)$, with $A : W \to V$ linear and $b \in V$.

For $x, y \in W$ and $t \in [0, 1]$, using linearity of A:

$$\begin{aligned} A(tx + (1-t)y) - b &= tAx + (1-t)Ay - b \\ &= tAx + (1-t)Ay - tb - (1-t)b \\ &= t(Ax - b) + (1-t)(Ay - b) \end{aligned}$$

So $A(\cdot) - b$ preserves convex combinations. Applying convexity of f_1:

$$\begin{aligned} g(tx + (1-t)y) &= f_1\big(A(tx + (1-t)y) - b\big) \\ &= f_1\big(t(Ax - b) + (1-t)(Ay - b)\big) \\ &\le tf_1(Ax - b) + (1-t)f_1(Ay - b) \\ &= tg(x) + (1-t)g(y) \end{aligned}$$

4 Problem 4

4.1 Task

Let $n = 1000$, $d = 3$, and define

$$a_i = \begin{pmatrix} 1 \\ t_i \\ t_i^2 \end{pmatrix}, \quad t_i = \frac{i}{n}, \quad x_{\text{true}} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

Set $b_i = a_i^\top x_{\text{true}}$ and consider

$$f(x) = \frac{1}{n} \sum_{i=1}^{n} (a_i^\top x - b_i)^2$$

(a) Show that

$$\nabla f(x) = \frac{2}{n}\sum_{i=1}^{n}(a_i^\top x - b_i)a_i$$

(b) Implement the following three methods with

$$x_0 = 0, \quad \eta = 0.01, \quad N = 50000, \quad m = 8$$

$$x_{k+1}^{\text{GD}} = x_k - \eta\nabla f(x_k) \tag{1}$$

$$x_{k+1}^{\text{SGD}} = x_k - \eta\, 2(a_{i_k}^T x_k - b_{ik})a_{i_k}, \quad i_k = 1 + (k \bmod n) \tag{2}$$

$$x_{k+1}^{\text{MB}} = x_k - \eta\,\frac{1}{m}\sum_{i\in B_k} 2(a_i^T x_k - b_i)a_i \tag{3}$$

where

$$B_k = \{1 + (km + j \bmod n) : j = 0, \ldots, m-1\}$$

(c) Plot $f(x_k)$ for all three methods.

4.2 Solution

4.2.1 (a)

$$\begin{aligned}
\nabla f(x) &= \frac{1}{n}\sum_{i=1}^{n}\nabla(a_i^\top x - b_i)^2 && \text{(linearity)}\\
&= \frac{1}{n}\sum_{i=1}^{n} 2(a_i^\top x - b_i)\,\nabla(a_i^\top x - b_i) && \text{(chain rule)}\\
&= \frac{2}{n}\sum_{i=1}^{n}(a_i^\top x - b_i)\,a_i,
\end{aligned}$$

since $\nabla_x(a_i^\top x - b_i) = a_i$ (b_i constant in x, and $\partial(a_i^\top x)/\partial x_j = (a_i)_j$).

4.2.2 (b) + (c)

```
# (b)
n = 1000
d = 3
```

```
eta = 0.01
N = 50_000
m = 8

t = [i/n for i in 1:n]
A = hcat(ones(n), t, t.^2)
x_true = [1.0, 2.0, 3.0]
b = A * x_true

f(x) = sum((A*x - b).^2) / n

function run_gd()
    x = zeros(d)
    hist = [f(x)]
    for k in 1:N
        grad = (2/n) * sum((A[i,:]'*x - b[i]) * A[i,:] for i in 1:n)
        x = x - eta * grad
        push!(hist, f(x))
    end
    hist
end

function run_sgd()
    x = zeros(d)
    hist = [f(x)]
    for k in 1:N
        i = 1 + (k % n)
        a_i = A[i, :]
        grad = 2 * (a_i' * x - b[i]) * a_i
        x = x - eta * grad
        push!(hist, f(x))
    end
    hist
end

function run_mb()
    x = zeros(d)
    hist = [f(x)]
    for k in 1:N
        indices = [1 + ((k*m + j) % n) for j in 0:m-1]
        A_batch = A[indices, :]
        b_batch = b[indices]
        grad = (2/m) * A_batch' * (A_batch*x - b_batch)
        x = x - eta * grad
        push!(hist, f(x))
    end
    hist
end
```

```
hist_gd  = run_gd()
hist_sgd = run_sgd()
hist_mb  = run_mb()

# (c)
using Plots
plot(0:N, hist_gd,  label="GD",  xlabel="iteration k", ylabel="f(x_k)",
     yscale=:log10, lw=2)
plot!(0:N, hist_sgd, label="SGD", lw=2)
plot!(0:N, hist_mb,  label="MB (m=8)", lw=2)

savefig("./assets/p_04_c.png")
```

5 Errors/Typos

- Problem 3: "be a normed spaces" → "be normed spaces" (number disagreement)
- Problem 4: equations (2) and (3) reference y_{i_k} and y_i, but the variable defined in the setup is $b_i = a_i^\top x_{\text{true}}$ — should be b_{i_k} and b_i respectively